
Airline Passenger Satisfaction Prediction

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Abstract

The Airline passenger satisfaction prediction attempts to predict whether a passenger is satisfied with their travel with a particular airline or are dissatisfied/neutral about their experience. This is a binary classification problem which aims to predict whether a customer is satisfied or not based on their responses to a survey. We also aim to identify the factors that are the most crucial in determining a passengers satisfaction.

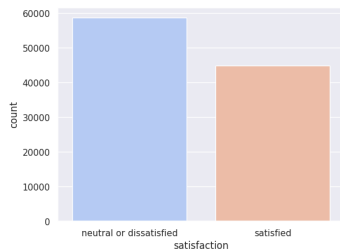
1 Problem Statement

The Airline Passenger Satisfaction Problem is a binary classification problem where we aim to classify whether a passenger would be satisfied or not based on how they rate various services offered by the airline on a scale of 1-5. We would be implementing various Machine Learning models to solve the binary classification, comparing their accuracy metrics and finding out which one comes out on top.

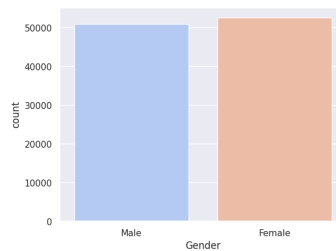
2 Data Description

The passenger survey data has been taken from **Kaggle** to perform binary classification.

- The training data consists of responses from 130,000 passengers with 25 feature columns.
- The testing data has responses from 26,000 passengers with 25 feature columns.
- The feature columns include quantitative data like id, gender and flight distance and categorical data like inflight entertainment, food and drinks, seat comfort, ease of online booking, baggage handling etc.
- The data is also balanced with about a 55:45 split among satisfied and neutral or dissatisfied individuals in the satisfaction feature that is to be predicted.



(a) Distribution of satisfied and unsatisfied classes



(b) Distribution of Gender

Figure 1: Data Distribution

3 Approach

The data was first cleaned by dropping the NA values and label encoding the categorical variables from strings to ordinal values. This is necessary because the models do not take string inputs and ordinal data helps the classifiers better understand the data. The Implementation steps are as follows:

3.1 Feature Selection

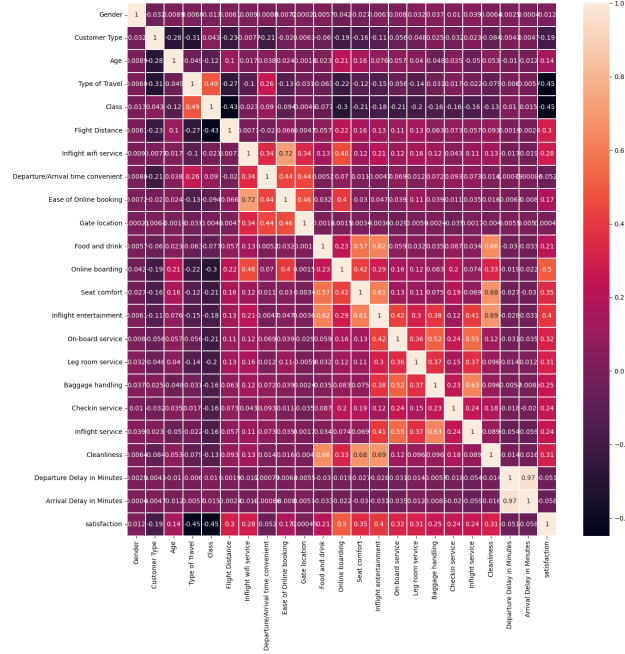


Figure 2: Heatmap

The First step was to select features so that attributes that do not contribute to the prediction can be pruned. For that we first plot a heatmap of the correlation of all the features in the dataset. The bottom row displays the correlation of the satisfaction attribute with all the other attributes. Departure delay in minutes, Arrival delay in minutes, Departure time convenience and gate location have correlation almost equal to 0. That means the satisfaction attribute is independent of the above listed attributes and therefore we can drop them. The features have also been scaled with a min max scaler. The min max scaler uses the following formula : $X_{scaled} = (X - X_{min}) / (X_{max} - X_{min})$. Scaling the data helps us normalize the input values and eliminate the preexisting biases.

3.2 ML Models

The following models were selected for classification since they all work very differently.

3.2.1 Baseline

We have also chosen a K-NN algorithm with no hyperparamter tuning and trained on raw data as our baseline. The accuracy came out to be 74%

3.2.2 Logistic Regression

Logistic regression was chosen for this task since it is perfect for binary classification. The model assigns weights to each feature attribute and uses the sigmoid activation function to map the input vectors. The model initially returned an accuracy of 83% but after dropping the independent features and normalization of attributes, the accuracy jumped to 87%

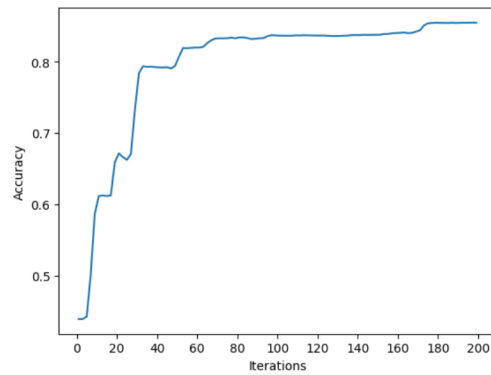


Figure 3: Accuracy over Iterations

3.2.3 K-Nearest Neighbors

The K Nearest Neighbours algorithm is another classification algorithm that calculates the class of an input data point based off of its N nearest neighbours classifications. The model initially recorded an accuracy of 75%. After normalizing data, its accuracy was boosted to 93%. This is because some of the features like flight distance, departure and arrival delay had a high numerical value and were skewing the distance calculations. K value set to 5 yielded the best results without overfitting.

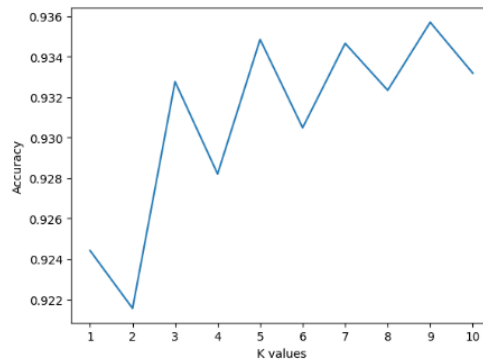


Figure 4: K values

3.2.4 Decision Tree

The decision tree can be used to classify which are the most important features that an airline can offer that make or break the customers opinion about their travel.

The decision tree was trained on the 17 features selected after feature selection. To find out what is the ideal max depth the tree should have, the tree was trained incrementally for 20 iterations.

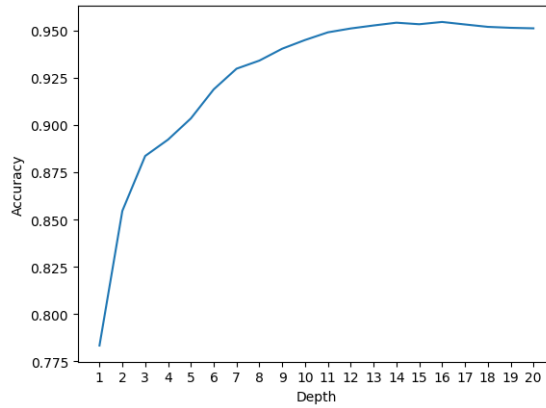


Figure 5: Decision tree accuracy vs max depth

So, the decision tree was trained with a max depth of 11 as the accuracy flattens out after this. If the max depth were to be increased it could lead to overfitting and the decision tree would perform worse on the test set. The decision tree was able to achieve an accuracy of 95%.

3.2.5 Neural Network

Finally, we used a neural network to perform the binary classification prediction on the dataset. The Neural network is developed using a Sequential model where the layers are defined one after the other.

The Neural network is made up of the following layers:

1. An initial input layer of dimension 17, which were selected for training after feature selection.
2. The second layer is a hidden layer Dense fully connected layer of 128 nodes where the fully connected layer implies that every node in the layer is connected to every node from the previous layer. A ReLU activation function was used in the hidden layer.
3. The third layer is also a Dense fully connected layer but with 64 nodes which also utilizes the ReLU activation function.
4. Lastly, the output layer consists of only 1 node as this is a binary classification problem where sigmoid activation function is used to predict where the customer is satisfied or not.

The Adam optimizer was used while compiling the model along with binary crossentropy as the loss. The neural network model gave us an accuracy of 95.53%.

```
Model: "sequential_1"
```

Layer (type)	Output Shape	Param #
dense_3 (Dense)	(None, 128)	2304
dense_4 (Dense)	(None, 64)	8256
dense_5 (Dense)	(None, 1)	65

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Total params: 10625 (41.50 KB)
Trainable params: 10625 (41.50 KB)
Non-trainable params: 0 (0.00 Byte)

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Figure 6: Model summary

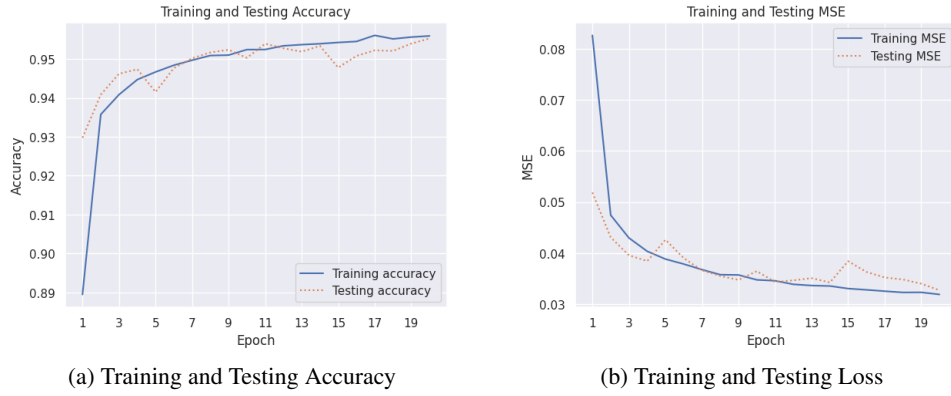


Figure 7: Accuracy and MSE loss of the Neural Network

4 Evaluation

The Machine Learning models were evaluated against a baseline model defined in Section 3.2.1 and the performance of the models is described in Table 1.

Table 1: Model Evaluation

Model	Accuracy	% Improvement Over Baseline
Logistic Regression	87%	17.5%
K - Nearest Neighbors	93%	25.5%
Decision Tree	94%	27%
Neural Network	95%	28%

5 Results

We identified the top 5 features with the most Information gain in the decision tree as the key factors that affect the passengers' opinion. These features are Online boarding, Inflight wifi service, Type of Travel, Inflight entertainment, Checkin Service. If the airlines focus on improving these aspects they can potentially increase customer satisfaction.